WEEKLY INTERNSHIP REPORT

Summer Research Programme 2025

Week Number	**Report Period**
Week 1 of 8	01 / 06 / 2026 → 06 / 06 / 2025
Group & Project Group 4 | Project Title: Event-Driven 6G Scheduling using SimPy + Sionna	**Mentor Name** Dr. Ch Anil Carie
Intern Name(s) 1. Snehitha (AP24110011008) 2. Shahistha Anjum (AP24110010993) 3. Shanmukh (AP24110011010) 4. Sandeep (AP24110010992)	**This week's title** Foundation Learning: Python, NetworkX, SimPy, Game Theory and Modbus Dataset Study

HOW THIS REPORT BUILDS YOUR CONFERENCE PAPER

Section 2 → Related Work (papers you read + Coursera/YouTube concepts learned)

Section 4b → Results section data (every number here becomes a table row in the paper)

Section 5 → Draft sentences copied directly into the paper manuscript

Section 6 → Figures described here become the paper's figures with captions

Section 8 → Limitations section and Future Work paragraph

Fill every section honestly. A blank section = missing paper content.

1 WEEK GOAL

Learn Python fundamentals, create a 5-node network topology using NetworkX, implement a basic event-driven simulation using SimPy, study Nash Equilibrium and Stackelberg Game concepts, and analyze the Modbus dataset to build the foundation for Event-Driven 6G Scheduling using SimPy and Sionna.

This week's goal: Build the foundational knowledge required for the Event-Driven 6G Scheduling project by learning Python programming, network modeling using NetworkX, event-driven simulation using SimPy, Game Theory concepts, and understanding the Modbus dataset structure.

Key deliverable (the one output that proves the goal was met):
Completed a 5-node network topology using NetworkX, developed a SimPy event simulation, prepared Nash Equilibrium and Stackelberg Game notes, and created a Modbus dataset summary for future scheduling experiments.

2 LEARNING — COURSERA + YOUTUBE

2A. Coursera Progress

Course Name	Modules Completed This Week	Score / Status
Python Programming Fundamentals	Variables, Loops, Functions, Lists, Dictionaries, File Handling	Completed
NetworkX Basics	Network Graphs, Nodes, Edges, Visualization	Completed
SimPy Basics	Environment, Process, Timeout, Event Simulation	Completed
Game Theory Fundamentals	Nash Equilibrium, Stackelberg Game	Completed
Modbus Dataset Study	Features, Attack Types, Dataset Structure	Completed

2B. YouTube Watched This Week

List every video watched. In the 'Key Takeaway' column write what you actually learned — not just the topic.

Video Title	Channel	Duration	Key Takeaway (1 sentence)
Python Full Course for Beginners	Programming with Mosh	6 hrs	Learned Python syntax, loops and functions

Video Title	Channel	Duration	Key Takeaway (1 sentence)
NetworkX Tutorial	Tech With Tim	30 min	Learned graph creation and visualization
SimPy Tutorial	Various	40 min	Learned event-driven simulation concepts
Game Theory Explained	Crash Course	20 min	Understood strategic decision making

2C. New Concept Learned (explain in your own words)

The most important concept learned this week was event-driven simulation using SimPy. In an event-driven system, actions occur when specific events happen rather than continuously. SimPy provides an environment where processes can be scheduled and executed based on simulation time. This concept is important because our project focuses on event-driven scheduling in future 6G networks. Understanding event scheduling will help in building realistic network simulations in upcoming weeks.

3 DAILY TASK LOG

Fill this DAILY — not on Friday from memory. One row per day.

Day	Task Completed	File / Output Committed to GitHub	Blocker (if any)
Mon	Installed Python, VS Code, Git, NetworkX and SimPy libraries. Explored the project objectives and internship roadmap.	Environment Setup Completed	Minor package installation issues resolved.
Tue	Learned Python fundamentals including variables, loops, functions, lists, dictionaries and file handling.	Python Practice Programs	None
Wed	Learned NetworkX concepts and created a 5-node network topology. Visualized the network using graph drawing functions.	network_topology.py	Graph layout adjustments required.
Thu	Learned SimPy fundamentals and implemented a simple event-driven simulation demonstrating packet transmission between devices.	simpy_demo.py	Understanding process scheduling required additional practice.
Fri	Studied Game Theory concepts including Nash Equilibrium and Stackelberg Games. Explored Modbus dataset structure and features.	Nash Notes, Stackelberg Notes, Dataset Summary	Understanding equilibrium concepts required extra reading.

Weekly Commit Summary

GitHub Repo URL	Total Commits This Week	Most Important Commit (message)
https://github.com/Snehithaa14/-RESEARCH-INTERN-WEEK---1-	5	Implemented 5-node network topology and basic SimPy event simulation.

4 TECHNICAL WORK THIS WEEK

4A. What I Built / Implemented

Developed a 5-node network topology using NetworkX where nodes represent devices and edges represent communication links. Implemented a basic event-driven simulation using SimPy to model packet transmission between devices over simulated time. Studied Game Theory concepts including Nash Equilibrium and Stackelberg Games and analyzed the Modbus dataset to understand its structure, features and attack categories. These activities established the foundation required for implementing Event-Driven 6G Scheduling in future weeks.

4B. Results & Numbers ★ MOST IMPORTANT ★

WHY THIS SECTION MATTERS

Every number you record here is a candidate row in your paper's results table.

RULE: If you ran an experiment this week, it MUST appear here as a number. No number = experiment not done.

These rows will be copied directly into the paper in Weeks 7–8.

Method / Model / Config	Metric 1 (name)	Metric 2 (name)	Metric 3 (name)	Notes
NetworkX Topology	Nodes = 5	Edges = 5	Connected Graph	Successfully Visualized
SimPy Simulation	Processes = 2	Events = 10	Sim Time = 20	Event-Driven
e.g. SVM (RBF)	Concepts = 5	Examples = 2	Notes Prepared	completed
Baseline from paper / prior work	Features = 20+	Attack Types = 3	Dataset Reviewed	Initial Analysis

Best result this week: Successfully developed a 5-node network topology and implemented a basic event-driven simulation using SimPy.

Compared to baseline: Week 1 focused on foundational learning and implementation. Performance evaluation will begin after scheduling models are developed.

4C. Key Figure / Plot This Week

file path: /results/network_topology.png

Caption draft: Figure 1 shows the 5-node network topology created using NetworkX. The graph represents communication links between devices and serves as the foundation for future 6G scheduling simulations.

5 PAPER CONTRIBUTION LOG

THIS SECTION IS THE HEART OF THE REPORT

Write these sentences THE SAME DAY as the experiment — not on Friday. Memory fades.

These sentences are copied into the paper in Weeks 7–8. The better you write here, the easier the paper sprint.

Every week must have at least 3 candidate sentences. Aim for 5.

5A. Candidate Sentences for the Paper

Format: [Section it belongs in] → [Draft sentence with numbers if possible]

Paper Section	Draft Sentence (write as it would appear in the paper)
Introduction	Future 6G networks require intelligent scheduling mechanisms capable of handling dynamic traffic demands and large-scale communication environments.
Related Work	Event-driven simulation frameworks are widely used for evaluating network scheduling algorithms under realistic operating conditions.
Methodology	A five-node network topology was developed using NetworkX to model communication between network entities.
Methodology	An event-driven simulation environment was implemented using SimPy to study process scheduling and communication events.
Results	Initial simulations successfully demonstrated event scheduling and process execution within the SimPy environment.

5B. Gap / Limitation Found This Week

The current implementation focuses only on foundational learning and small-scale simulations. Advanced scheduling algorithms, wireless channel modeling and Sionna integration have not yet been implemented. These components will be explored in future weeks to develop a realistic 6G scheduling framework.

5C. Reference Found This Week

#	IEEE Citation (Author, Title, Journal/Conf, Year)	Relevance to Our Work
1	A. A. Hagberg, D. A. Schult, and P. J. Swart, “Exploring Network Structure, Dynamics, and Function using NetworkX,” Proc. 7th Python in Science Conference, 2008.	Helped understand network graph modeling and visualization.
2	T. S. Rappaport et al., “Wireless Communications and Applications Above 100 GHz: Opportunities and Challenges for 6G and Beyond,” IEEE Access, 2019.	Introduced key concepts and requirements of future 6G networks.

6 NEXT WEEK PLAN

Plan this before you leave on Friday. Specific tasks only — not 'continue work on the model'.

Day	Planned Task (specific)	Expected Output / Commit
Mon	Study Graph Theory concepts including Degree Centrality, Betweenness Centrality, Closeness Centrality, and Network Density.	graph_theory_notes.pdf committed with examples and formulas.
Tue	Develop the create_iot_network() function and generate a 50-node IoT network using NetworkX.	iot_network_generator.py committed with 50-node topology visualization.
Wed	Study IoT security threats including DoS, DDoS, Reconnaissance, Malware, and Botnets.	iot_security_notes.md committed with attack descriptions and examples.
Thu	Implement the attack_node() function and simulate Random, Targeted, and Burst attacks on the network.	attack_engine.py committed with attack logs and statistics.
Fri (demo)	Implement the monitor_node() defender function, generate attack statistics, network statistics, and visualize network behavior. Demonstrate the complete baseline system to mentor.	defender_engine.py, attack_logs.csv, network_visualization.png, and Week 2 report to be submitted.

Coursera + YouTube Plan for Next Week

Platform	Module / Video Title	When to watch / complete
YouTube	Graph Theory Explained	Monday before coding
YouTube	NetworkX Centrality Measures Tutorial	Tuesday before implementation
YouTube	IoT Security Attacks Explained (DoS, DDoS, Botnets)	Wednesday evening
YouTube	Network Attack Simulation using Python	Thursday before attack engine development

7 SELF-ASSESSMENT

WHY THIS SECTION EXISTS

These two questions build the skills that separate researchers from coders.

Q1 answers become the 'Surprising Finding' paragraph in the paper's Discussion section.

Q2 answers become better experimental design next week — and better research judgment overall.

Answer honestly. 3–5 sentences each. This is for your growth, not for marking.

Q1: What surprised you most technically this week?

I was surprised by how effectively SimPy can model real-world systems using simple event scheduling concepts. Initially, I expected network simulations to require complex implementations, but SimPy allowed processes and events to be created with relatively simple code. I also found that NetworkX could represent network structures visually using only nodes and edges. This showed me how powerful simulation and graph modeling tools can be when designing large-scale communication systems such as future 6G networks.

Q2: If you had to redo this week, what would you do differently?

If I had to redo this week, I would spend more time implementing practical examples immediately after learning each concept. For example, after learning NetworkX, I would experiment with larger network topologies instead of only creating a small 5-node network. Similarly, after learning SimPy, I would test different event scheduling scenarios to better understand process interactions. This approach would provide deeper practical understanding and prepare me more effectively for the larger simulations planned in Week 2.

8 MENTOR REVIEW (Mentor fills this section)

Students submit the report by Friday 5pm. Mentor completes this section by Monday 9am before the next week begins.

8A. Checklist

Item	Status	Comment
Key deliverable submitted and committed to GitHub		
Section 4B results table has real numbers		
Section 5A has ≥ 3 candidate paper sentences		
GitHub commits are clean with meaningful messages		
Friday demo completed		
Results are reproducible (someone else can run the code)		

8B. Technical Feedback

8C. Paper Progress Assessment

Paper Section	Status After This Week	Target Week to Complete
Introduction	Not started / Outline / Draft / Final	Week 7
Related Work	Not started / Outline / Draft / Final	Week 6
Methodology	Not started / Outline / Draft / Final	Week 7
Results	Not started / Data collected / Draft / Final	Week 7
Conclusion	Not started / Draft / Final	Week 7
References	Not started / Partial (__ refs) / Complete	Week 8

8D. Go / No-Go for Next Week

✅ **PROCEED**	⚠ **FIX FIRST**	🔴 **REPLAN**
All deliverables met. Continue to next week's plan.	Fix specific items before proceeding. See feedback above.	Results not credible. Stop and redesign the experiment.

Decision: Circle one above. If FIX FIRST — list exact items to fix before Monday:

8E. Mentor Signature

Mentor Name	Date Reviewed	Signature

APPENDIX — CUMULATIVE PAPER SKELETON TRACKER

Update this page every week. By Week 6 this should be the first draft of your paper.

HOW TO USE THIS TRACKER

Each row corresponds to a paper section. Paste your best candidate sentences from Section 5A here each week.

By Week 6 you should have 3–5 sentences per row. These become the paper draft in Week 7.

Colour code: Black = confirmed content Grey = draft Red = needs rewrite

A. Introduction

Added in	Content / Draft Sentences
Week 1	
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	

B. Related Work

Added in	Content / Draft Sentences
Week 1	
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	

Week 7	
Week 8	

C. Methodology

Added in	Content / Draft Sentences
Week 1	
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	

D. Results

Added in	Content / Draft Sentences
Week 1	
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	

E. Discussion / Conclusion

Added in	Content / Draft Sentences
Week 1	

Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	

F. References (IEEE format)

Added in	Content / Draft Sentences
Week 1	
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	